

THE WHOLE OF HIGHER APPLICATIONS OF MATHS 2026 In 4 HOURS!

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**LAST MINUTE HIGHER APPS LIVE STREAM
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4. Some helpful R commands

Entering data to R Studio

To read in data from an Excel csv file called `excel_data.csv` to R Studio and name it `mydata`, first use the drop down menus in R Studio **Session > Set Working Directory > Choose Directory** to indicate the location of `excel_data.csv` on your computer. The following code will then read the data into R Studio:

```
mydata<-read.csv("excel_data.csv")
```

`attach(mydata)` — this adds the variable names

At the end of the analysis remember to use `detach(mydata)` to disassociate the variable names.

(a) Graphics

If you have the numeric variables X and Y:

`hist(X, main= "Title",xlab="x-axis label",ylab="Frequency")` — this produces a histogram of the variable named X, it adds a title and axis labels

`boxplot(Y, main="Title", ylab="y-axis label")` — produces a boxplot of the numerical variable Y

`boxplot(X,Y, main="Title", xlab="x-axis label", ylab="y-axis label", names=c("X", "Y"))` — produces a comparative boxplot of the numerical variables X and Y

`plot(X,Y, main="Scatterplot of Y on X",xlab="x-axis label",ylab="y-axis label")` — produces a scatterplot of Y on X

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If you have the categorical variable X:

`table(X)` — computes the number of observations in each level of the categorical variable X

`pie(table(X), main="Title")` — this gives a simple pie chart of the categories in variable X with the specified title

`barplot(table(X), main="Title", xlab="x-axis label", ylab="Frequency")` — this gives a bar chart of the categorical variable X with the required title and axis labels

(b) Descriptive Statistics

`mean(X)` — computes the mean of the numerical variable X

`sd(X)` — computes the standard deviation of the numerical variable X

`summary(X)` — computes the mean, median, minimum, maximum and upper and lower quartiles of the numerical variable X

`IQR(X)` — computes the interquartile range of the numerical variable X

`prop.table(table(X))` — returns the proportion of observations in each level of the categorical variable X

`prop.table(table(X))*100` — returns the percentage of observations in each level of the categorical variable X

`table(X,Y)` — produces a cross-tabulation between the two categorical variables X and Y

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(c) Correlation and Regression

`cor.test(X,Y)` — computes the correlation between X and Y and performs a test of the null hypothesis of zero correlation

`lm(Y~X)` — fits a linear regression line to the data (lm command stands for linear model)

`abline(lm(Y~X))` — adds the least squares linear regression line to an existing scatterplot of Y on X

`summary(lm(Y~X))` — displays the coefficient of determination (R-squared)

To predict with your Linear Model:

`predict(lm(Y ~ X), newdata=data.frame(X=C), interval = "pred")` — computes the predicted value of Y when X=C along with a 95% prediction interval

(d) Hypothesis Testing

`t.test(X,Y)` — performs a two-sample t-test between X and Y

`t.test(X,Y,paired=TRUE)` — performs a paired t-test between X and Y

`prop.test(x = c(a, b), n = c(n1, n2))` — performs a two-sample test for equality of proportions

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HIGHER APPLICATIONS OF MATHS FINANCE UNIT REVISION

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FINANCE – LOAN SCHEDULE WRITTEN



Bailey takes out a loan for £4000 with an annual effective rate of interest of 29.9%.

(a) Calculate the monthly effective rate of interest.

1

Bailey makes level monthly repayments of £250 at the end of each month.

(b) Complete the following loan schedule for Bailey's loan to show the loan outstanding at the end of month 2.

2

Time (months)	Repayment (£)	Interest content of repayment (£)	Capital content of repayment (£)	Loan outstanding (£)
0				4000.00
1	250.00			
2	250.00			

1. Jill borrows £5650 from a bank.

The annual effective rate of interest on the loan is 9%.

Jill makes level monthly repayments of £186.01 at the end of each month.

Complete the following loan schedule for Jill's loan to show the loan outstanding at the end of month 2.

3

Time (months)	Repayment (£)	Interest content of repayment (£)	Capital content of repayment (£)	Loan outstanding (£)
0				5650
1	186.01			
2	186.01			

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FINANCE – REPAYMENT SCHEDULE WRITTEN



Freddie has a credit card with an annual effective rate of interest of 29.9%. Interest is applied at the end of each month.

Payments must be made on the first day of each month. The minimum payment must be either 5% of the balance outstanding at that time or £5, whichever is higher.

After Freddie makes his payment on 1 March, the balance of his credit card was £823.19.

Freddie does not use his credit card during March.

- (a) Calculate the balance of his credit card on 1 April after Freddie has made the minimum payment. 3
- (b) Give one reason why Freddie should consider paying the full balance of his credit card each month. 1

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FINANCE ACCUMULATION (TIMELINE) & PRESENT VALUE

You must refer to the information on 'Lifetime ISA' given in the pre-release material when answering this question.

Morgan opens a Lifetime ISA with an online bank. The account has an annual effective rate of interest of 3.25%.

Morgan makes a deposit of £2500 into the account on 1 January 2024. The government bonus is paid into her account on the first day of the following month. She makes no further deposits in 2024.

Calculate the accumulated value of Morgan's Lifetime ISA on 31 December 2024.

4

2. Lifetime ISA

You can use a Lifetime ISA (Individual Savings Account) to buy your first home or save for later life. You must be 18 or over but under 40 to open a Lifetime ISA.

You must make your first deposit into your ISA before your 40th birthday and can make further deposits until the day before your 50th birthday.

You can save up to £4000 a year as a lump sum or by putting in cash when you can. The government will add a 25% bonus to your savings, up to a maximum of £1000 per year.

The bonus is paid every month, on new deposits that have been made.

Once in your account the bonus becomes part of your savings and interest can be applied.

When you turn 50, you will not be able to make any further deposits into your Lifetime ISA or earn any further bonuses. Your account will stay open, and your savings will still earn interest.

You can withdraw money from your ISA if you are:

- buying your first home
- aged 60 or over
- terminally ill, with less than 12 months to live.

You will pay a withdrawal charge of 25% if you withdraw cash or assets for any other reason (also known as making an 'unauthorised withdrawal'). This charge is deducted from the amount withdrawn. This recovers the government bonus you received on your original deposits.

Adapted from: www.gov.uk/lifetime-isa

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FINANCE ACCUMULATION (TIMELINE) & PRESENT VALUE

5. Ewa deposited £4500 in a variable rate savings account on 1 January 2021.

The effective rates of interest for the savings account are as follows:

Dates	Interest rate
1 January 2021 to 31 March 2021	0.415% per month
1 April 2021 to 31 December 2021	4.7% per year
From 1 January 2022	2.6% per year

- (a) Calculate Ewa's balance on 1 January 2024.

3

On 1 January 2022 Blair opened an account with the same effective rates of interest. Blair has a savings goal of £6000 by 1 January 2024.

- (b) Calculate the minimum deposit Blair should have made on 1 January 2022 to achieve this savings goal.

1

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FINANCE – ACCUMULATION (TIMELINE) & PRESENT VALUE (FINDING ERI)

Taylor opens a savings account on 1 April 2021 with an initial deposit of £400.

The effective rates of interest for the savings account are as follows.

Dates	Interest rate
1 April 2021 to 31 January 2022	1.2% per year
1 February 2022 to 31 July 2022	0.11% per month
From 1 August 2022	1.7% per year

Taylor makes further deposits of £200 on 1 August 2021 and £250 on 1 April 2022.

(a) Calculate the balance in Taylor's savings account on 31 July 2022.

4

Taylor made a **final** deposit into this account on 1 August 2022.

On 1 December 2022, their savings account balance was £1000.

(b) Calculate how much they deposited on 1 August 2022.

2

FINANCE – OVERALL ERI & PRESENT VALUE

6. The average price of petrol increased at the following annual effective rates:

- March 2018 to 2019: 2.1%
- March 2019 to 2020: 0.5%
- March 2020 to 2021: 2.0%

(a) Calculate the overall percentage increase in the average price of petrol over the three years from March 2018 to March 2021.

1

The average price of petrol in March 2021 was 136.4 pence per litre.

(b) Hence calculate the average cost of filling a 45-litre tank with petrol in March 2018.

2

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FINANCE – CONSUMER PRICE INDEX

9. The Consumer Price Index (CPI) in the UK in April 2021 was 110.4, relative to a baseline of 100 in April 2015.

(a) Explain what this figure means in terms of relative purchasing power.

1

The price of a new 3-door car rose in line with CPI between April 2015 and April 2021.

In 2021, the price of a new 3-door car was £14,108.

(b) Calculate the price of a new 3-door car in April 2015.

1

The National Living Wage is the minimum hourly rate to be paid to any employee aged 23 or older.

In April 2021, the National Living Wage was £8.91. This was raised to £9.50 in April 2022.

(c) Given that the CPI in April 2022 was 119.0, determine whether this rise in the National Living Wage was in line with inflation.

2

The Consumer Price Index (CPI) in April 2022 was 119.0 and in April 2023 was 128.3.

In April 2022, Tom's gross monthly salary was £4200.

In April 2023, his gross monthly salary was increased to £4500.

(b) Determine whether Tom's gross monthly salary increased in line with the CPI.

2

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3. You must refer to the information on ‘Deductions from salaries’ given in the pre-release material when answering this question.

Alaina earns a gross annual salary of £24,960. She pays 8.2% of her gross annual salary into her pension and pays £82.56 National Insurance per month.

Calculate her **net monthly salary** after all deductions.

4

1. Deductions from salaries

Scottish tax bands 2024/25

Band	Taxable income	Scottish tax rate
Personal Allowance	Up to £12,570	0%
Starter rate	£12,570 — £14,876	19%
Basic rate	£14,876 — £26,561	20%
Intermediate rate	£26,561 — £43,662	21%
Higher rate	£43,662 — £75,000	42%
Advanced rate	£75,000 — £125,140	45%
Top rate	Over £125,140	48%

National Insurance contributions

You begin paying National Insurance once you earn more than £1048 a month (this is the amount for the 2024/25 tax year).

For payslips dated between 6 April 2024 and 5 April 2025, you pay 8% of your monthly earnings between £1048 and £4189; 2% of your monthly earnings above £4189.

National Insurance is calculated on a person’s salary **before deductions** such as pension contributions.

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You must refer to the information on ‘Deductions from salaries’ given in the pre-release material when answering this question.

In the 2023/24 tax year, Tom was paid a gross salary of £4500 per month for 12 months, from the 6 April 2023.

His annual income tax deduction is £10,718.48.

Tom has opted out of paying any pension contributions.

(a) Calculate Tom’s net annual salary for the 2023/24 tax year, after all deductions including National Insurance.

4

1. Deductions from salaries

Scottish income tax bands 2023/24

Band	Taxable income	Scottish tax rate
Personal Allowance	Up to £12,570	0%
Starter rate	£12,570 – £14,732	19%
Basic rate	£14,732 – £25,688	20%
Intermediate rate	£25,688 – £43,662	21%
Higher rate	£43,662 – £125,140	42%
Top rate	Over £125,140	47%

National Insurance Contributions

In the 2023/24 tax year, National Insurance payments varied during the year.

You begin paying National Insurance once you earn more than £1048 a month (this is the amount for the 2023/24 tax year).

For payslips dated between 6 April 2023 and 5 January 2024, you pay 12% of your monthly earnings between £1048 and £4189; 2% of your monthly earnings above £4189.

For payslips dated between 6 January 2024 and 5 April 2024, you pay 10% of your monthly earnings between £1048 and £4189; 2% of your monthly earnings above £4189.

National Insurance is calculated on a person’s salary **before deductions** such as pension contributions.

4. You must refer to the information on 'Scottish Tax Bands 2022/23' given in the pre-release material when answering this question.

James works for a large building company. He is a permanent employee and earns a gross annual salary of £36,700.

James pays 4.7% of his gross annual salary into his pension and pays £3071.45 National Insurance per annum.

Calculate his **net annual salary** after all deductions.

4

1. Deductions from salaries

Scottish tax bands 2022/23

Band	Taxable income	Scottish tax rate
Personal Allowance	Up to £12,570	0%
Starter rate	£12,570 – £14,732	19%
Basic rate	£14,732 – £25,688	20%
Intermediate rate	£25,688 – £43,662	21%
Higher rate	£43,662 – £150,000	41%
Top rate	Over £150,000	46%

National Insurance Contributions

In the 2022–23 tax year, National Insurance payments varied during the year.

For payslips dated between 6 April 2022 and 5 July 2022, you began paying National Insurance once you earned more than £823 a month. The National Insurance rate you paid depended on how much you earned: 13.25% of your monthly earnings between £823 and £4189; 3.25% of your monthly earnings above £4189.

For payslips dated between 6 July 2022 and 5 November 2022, you began paying National Insurance once you earned more than £1048 a month. The National Insurance rate you paid depended on how much you earned: 13.25% of your monthly earnings between £1048 and £4189; 3.25% of your monthly earnings above £4189.

For payslips dated between 6 November 2022 and 5 April 2023, you began paying National Insurance once you earned more than £1048 a month. The National Insurance rate you paid depended on how much you earned: 12% of your monthly earnings between £1048 and £4189; 2% of your monthly earnings above £4189.

National Insurance is calculated on a person's salary **before** deductions such as pension contributions.

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SPREADSHEET
FILE LINK

You must refer to the spreadsheet file 'Q8 Finance.xlsx' when answering this question.

You must complete parts (a) and (c) using the spreadsheet file.

Part (b) must be completed in the answer space provided.

Martin is buying a car for £7500. The garage offered Martin the finance package below:

- 20% deposit
- 35 level monthly repayments of £216.77
- final monthly repayment £216.13.

- (a) Open the 'Car Finance' worksheet.
- Complete the loan schedule to determine the annual effective rate of interest that is being charged for the finance package. 5

Car insurance is a legal requirement for cars driven on UK roads.

- (b) Explain the purpose of car insurance. 1

Martin has been saving for the deposit and car insurance. He opened a savings account with an initial deposit of £500 on 1 August 2023.

He made regular monthly payments of £250 into the account on the first day of each month from 1 September 2023. The effective rates of interest for the savings account are as follows.

Dates	Effective rate of interest
1 August 2023 to 30 November 2023	0.24% per month
From 1 December 2023	3.1% per year

- (c) Open the 'Savings' worksheet.
- Calculate the balance in Martin's savings account immediately before he makes his payment on 1 August 2024. 4

Print the 'Car Finance' worksheet in value view and in formula view.

Print the 'Savings' worksheet in value view and in formula view.



Esme is building an extension to her house.

She has been offered a £25,000 mortgage with an effective annual rate of interest of 3.5% over 5 years.

Open the ‘Mortgage’ worksheet.

- (a) Complete the ‘Mortgage schedule’ to determine the level monthly repayment amount, and the final repayment amount. 4

Esme needs new building insurance once the extension is completed.

She is choosing between the following two options.

Cost per year(£)	Total excess(£)
216.94	350
281.95	100

- (b) (i) State one advantage of having a high excess amount on your insurance policy. 1

One of the windows in Esme’s house is broken and she has to decide whether to make a claim.

- (ii) Explain why Esme may choose not to make a claim using her insurance policy. 1

The maximum monthly repayment allowed by the lender is £550.

Esme chooses to reduce the term of her mortgage by increasing her monthly repayments to the maximum amount.

Open the ‘Increased payments’ worksheet.

- (c) (i) Complete the ‘Increased payments schedule’ for the reduced term and calculate the final repayment amount. 2
- (ii) Determine how much money this would save Esme over the term of her mortgage. 1

Print the ‘Mortgage’ worksheet in value view and in formula view.

Print the ‘Increased payments’ worksheet in value view and in formula view.



You must refer to the spreadsheet file ‘Q11 Ramsay’s Loan.xlsx’ when answering this question.

You must complete parts (a) (i), (a) (ii), (c) (i), and (c) (ii) using the spreadsheet file.

Part (b) must be completed in the space provided.

Ramsay applies to take out a loan of £6000 with a term of 3 years from a bank.

Level monthly repayments are made at the end of each month.

The effective annual interest rate is 6.3%.

Open the ‘Bank Loan’ worksheet.

- | | | | |
|-----|------|--|---|
| (a) | (i) | Complete the ‘Bank loan repayment schedule’ to determine the level monthly repayment amount, and the final repayment amount. | 4 |
| | (ii) | Determine the total interest paid over the term of the loan. | 1 |

Ramsay’s application for this loan was rejected.

- | | | |
|-----|--|---|
| (b) | State one reason why a bank might reject a loan application. | 1 |
|-----|--|---|

Ramsay decides to borrow the money from a loan company.

The loan company offers Ramsay £6000 for 3 years with fixed level monthly repayments of £250 made at the end of each month.

Open the ‘Loan Company’ worksheet.

- | | | | |
|-----|------|---|---|
| (c) | (i) | Complete the ‘Loan company repayment schedule’ to find the annual effective rate of interest. | 2 |
| | (ii) | Determine the difference in total interest paid (in £) between the two loans. | 1 |

Print the ‘Bank Loan’ worksheet in value view and formula view. Print the ‘Loan Company’ worksheet in value view and formula view.



9. You must refer to the spreadsheet file 'Q9 Car Repayments.xlsx' when answering this question.

You must complete parts (a), and (b) using the spreadsheet file.

Part (c) must be completed in the space provided.

Maria is buying a car for £15,000. She has arranged a loan for the full amount from the bank, to be paid back with level monthly repayments for 48 months. The annual effective interest rate is 9.5%.

Open the 'Bank Loan' worksheet.

- (a) Complete the 'Loan Repayment Schedule' to determine the level monthly repayment amount, and the final repayment amount.

4

The car dealership has their own finance options.

Option 1

- Pay £300 per month for 48 months, and return the car to the dealer.
- Additional charges will apply if the mileage exceeds 24 000 miles when the car is returned.

Option 2

- Pay £300 per month for 48 months, and keep the car by making an additional payment of £5000 in the final month.

Open the 'Car Finance' worksheet.

- (b) Complete the 'Finance Repayment Schedule' to find the annual effective interest rate of Option 2.

3

- (c) State two reasons why Maria might decide to purchase the car using the bank loan instead of the car dealership finance options.

2

Print the 'Bank Loan' worksheet in value view and in formula view. Print the 'Car Finance' worksheet in value view and in formula view.

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10. You must refer to the spreadsheet file 'Q10 Carol's Gift' when answering this question. You must complete parts (b), (c) (i) and (c) (ii), using the spreadsheet file. Parts (a) and (d) must be completed in the answer spaces provided.

Carol has received a gift of £2500, and is considering what to do with it.

She has a savings account that pays interest at an annual effective rate of 1.25%.

- (a) Calculate how much **interest** Carol would earn if she invested this gift in her savings account for 34 months.

2

Carol also has a personal loan. She originally borrowed £8000 to be repaid by level monthly repayments for 48 months, with the first repayment made one month after she took out the loan. Interest is charged at an annual effective rate of 4.9%.

- (b) Open the 'Original Loan' worksheet. Complete formulae in the loan schedule and calculate the level monthly repayment amount, and the final repayment amount.

4

Carol has just made the 14th monthly repayment on the loan. She decides to find out the impact of using the £2500 gift as a lump sum payment to reduce the outstanding balance on her loan.

The loan provider agrees to recalculate a new level monthly repayment amount, to be paid in each of the remaining 34 months.

- (c) (i) Copy the 'Original Loan' worksheet. Rename the copy to 'Pay Lump Sum'. Adjust the 'Pay Lump Sum' worksheet as required, and hence calculate Carol's new level monthly repayment.
- (ii) On the 'Pay Lump Sum' worksheet, calculate how much Carol would save in interest payments by making this lump sum payment.

3

2

- (d) State one reason why Carol might choose to pay the gift into her savings account, rather than use it to reduce the balance on her loan.

1

Print your answers to Q10 (b), (c) (i) and (c) (ii) in:

- value view
- formula view.

HIGHER APPLICATIONS OF MATHS DECISION AND PLANNING MAKING UNIT REVISION

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PERT AND GANTT CHARTS

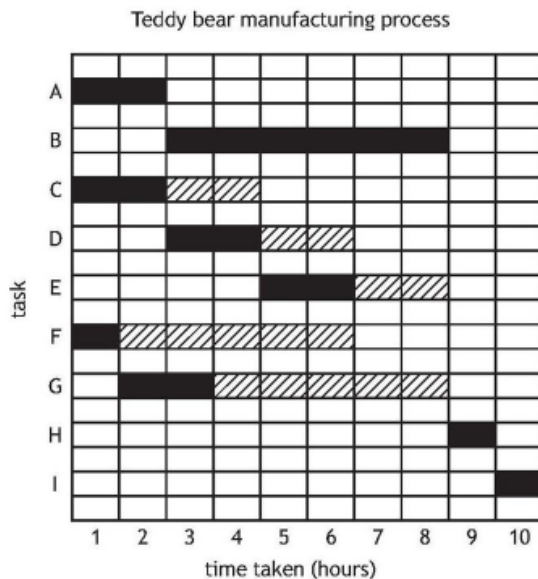
A company makes hand-made teddy bears.

The tasks required to make each teddy bear are listed below.

	Task
A	Cut fur
B	Stuff and sew fur
C	Cut material
D	Sew clothes
E	Embroider t-shirt
F	Cut accessories
G	Sew accessories
H	Dress and accessorise
I	Package and ship bear

The Gantt chart shows the tasks, with float times, when three employees are available to make each teddy bear.

Each task is completed by a **single** employee.



Float times in the Gantt chart are represented by .

(a) State the critical path.

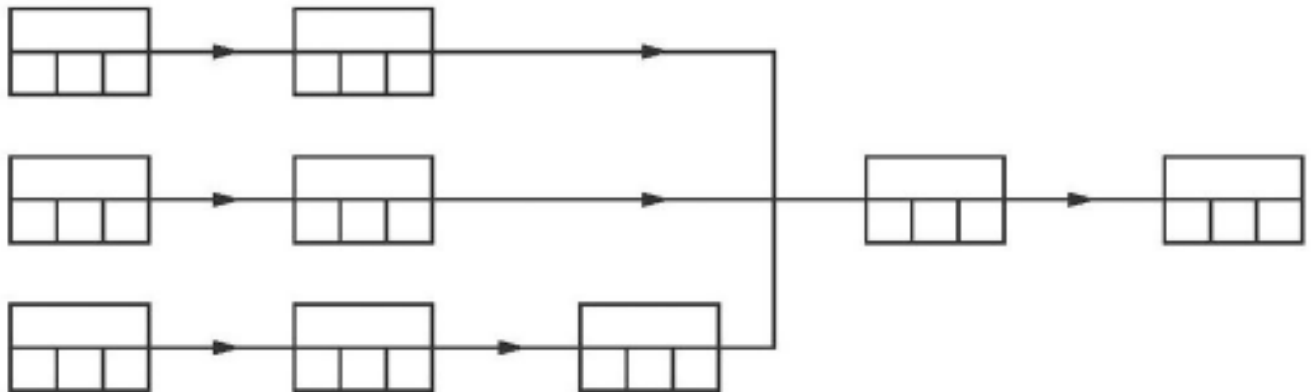
1

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- (b) Complete the PERT chart showing the earliest start time and the latest completion time for each task.

4



- (c) Determine the minimum time taken to make each teddy bear when only two employees are available.

1

PERT AND GANT CHARTS

3. A ship is being repaired.

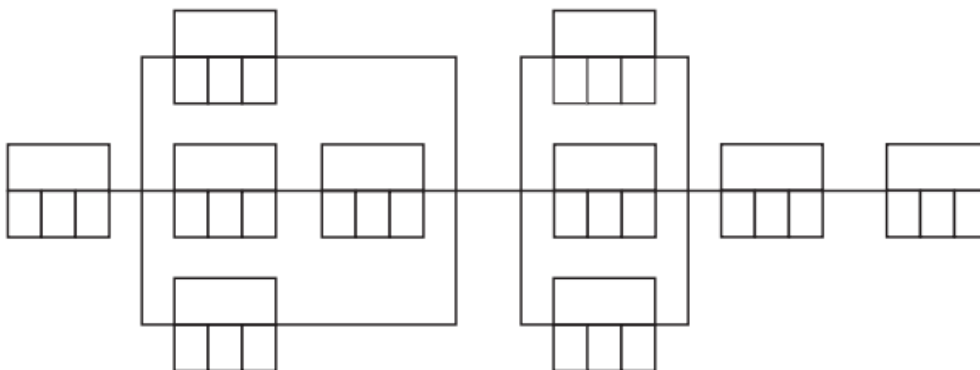
The propeller must be removed for the repair to be carried out.

The table shows the list of tasks required to complete this job, the time required for each task, and the order in which the tasks must be completed.

Task	Description	Preceding task	Duration (hours)
A	Disconnect electrics	none	2
B	Remove oils from system	A	12
C	Remove rudder	A	24
D	Disconnect oil pipes and hoses	A	8
E	Prepare oil pipes and hoses to be reconnected	D	4
F	Remove propeller blades	B, C, E	14
G	Disconnect propeller stern seals	B, C, E	12
H	Disconnect propeller shaft coupling	B, C, E	8
I	Remove propeller shaft coupling and internal pipes	F, G, H	6
J	Take out propeller shaft	I	8

(a) Complete the PERT chart showing the earliest start time and the latest completion time for each task.

5



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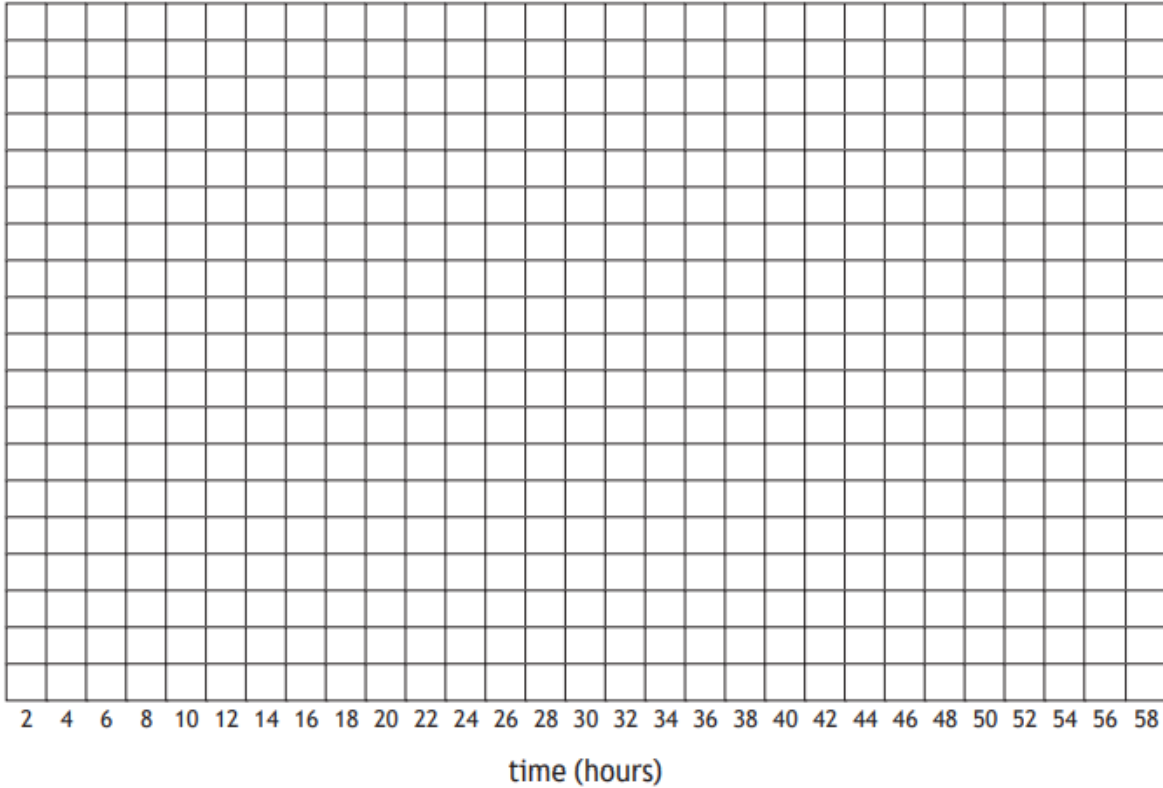
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PERT AND GANTT CHARTS

(b) Construct a Gantt chart without float times for this job.

3

(An additional diagram, if required, can be found on page 21.)



(c) Determine the critical path and the minimum time required to complete the job.

2

During the job there are difficulties disconnecting the propeller shaft coupling (task H).

(d) Determine the maximum time that can be taken to disconnect the propeller shaft coupling without delaying the total completion time of the job.

1

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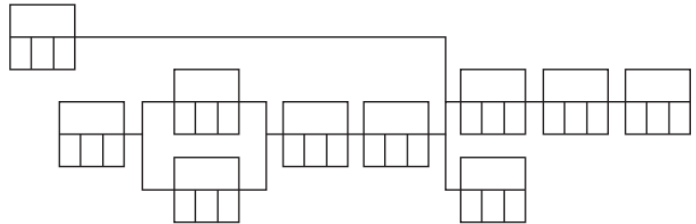
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PERT AND GANT CHARTS

2. In a cooking competition, a contestant has 80 minutes to make a crème brûlée. There are several tasks that need to be completed to make the crème brûlée. The contestant's timings for each task are shown.

Task	Description	Preceding task	Time (minutes)
A	heat oven	none	8
B	gather ingredients and prepare bowls	none	10
C	whisk egg yolks and sugar	B	5
D	heat double cream in pan	B	4
E	add cream to egg yolk mixture	C, D	2
F	sieve mixture and put into bowls	E	4
G	put bowls into oven and cook	A, F	35
H	tidy workspace	F	12
I	sprinkle caster sugar over each bowl and caramelise	G	3
J	plate up and serve	I	5

- (a) Complete the PERT chart below to allow the contestant to determine the earliest start time and latest completion time for each task.
(An additional diagram, if required, can be found on page 16.)



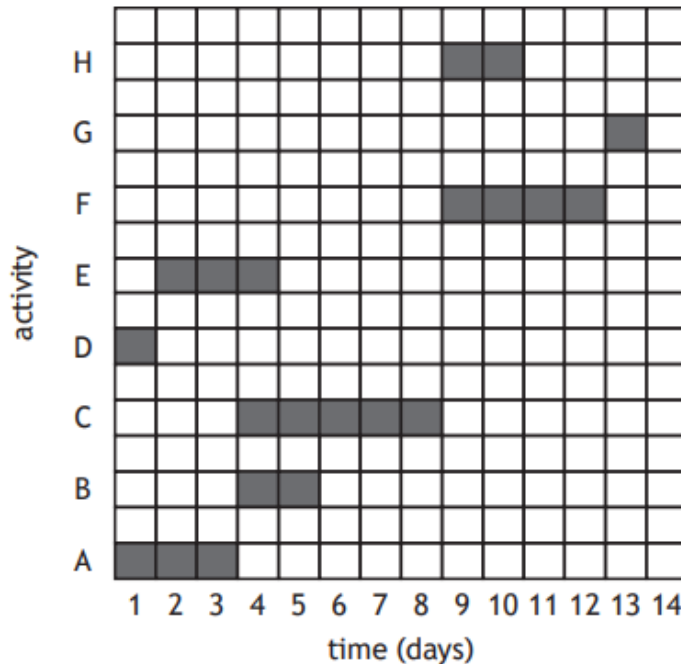
When the contestant puts the bowls into the oven, they realise the oven has not been heated.

- (b) Determine whether they can still finish on time.
Give a reason for your answer.

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6. The Gantt chart below shows the activities required to complete a project.



Activity G is dependent on activities F and H.

Activity F is dependent on activities B, C and E.

Activity E is dependent on activity D.

Activities C and B are dependent on activity A.

(a) State the critical path for this project.

1

(b) State the duration of the float time for activity B.

1

(c) State one advantage of using a Gantt chart.

1

PERT & GANT CHARTS

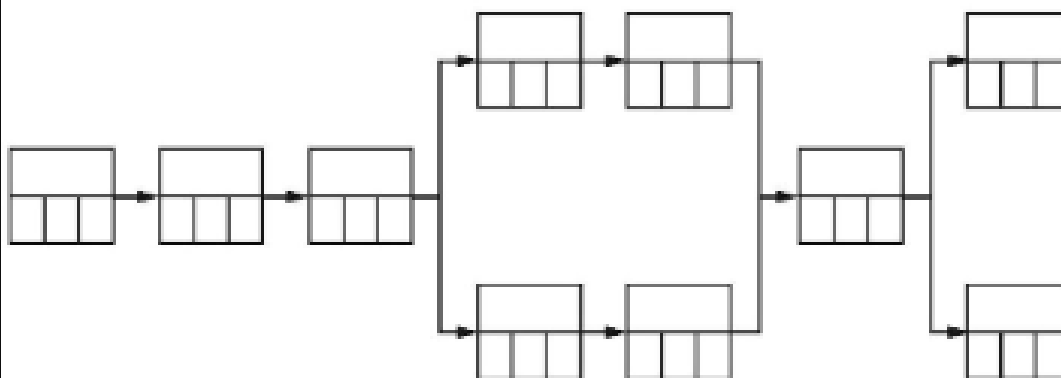
A company provides a team of workers to re-roof a house.

The table shows the list of tasks required to complete this job, the time required for each task, and the order in which the tasks are to be completed.

Task	Description	Preceding task	Time (hours)
A	Transport materials to location	None	3
B	Construct scaffolding	A	7
C	Remove old tiles	B	4
D	Take old tiles to recycling site	C	1
E	Inspect roofing	C	2
F	Purchase required number of roof tiles	D	1
G	Replace roofing felt	E	4
H	Replace roofing tiles	F, G	6
I	Remove scaffolding	H	3
J	Clean up	H	2

- (a) Complete the PERT chart showing the earliest start time and the latest completion time for each task.

6



- (b) The roofing company works for 9 hours each day on this job.

State the minimum number of days that the company will require to complete this job.

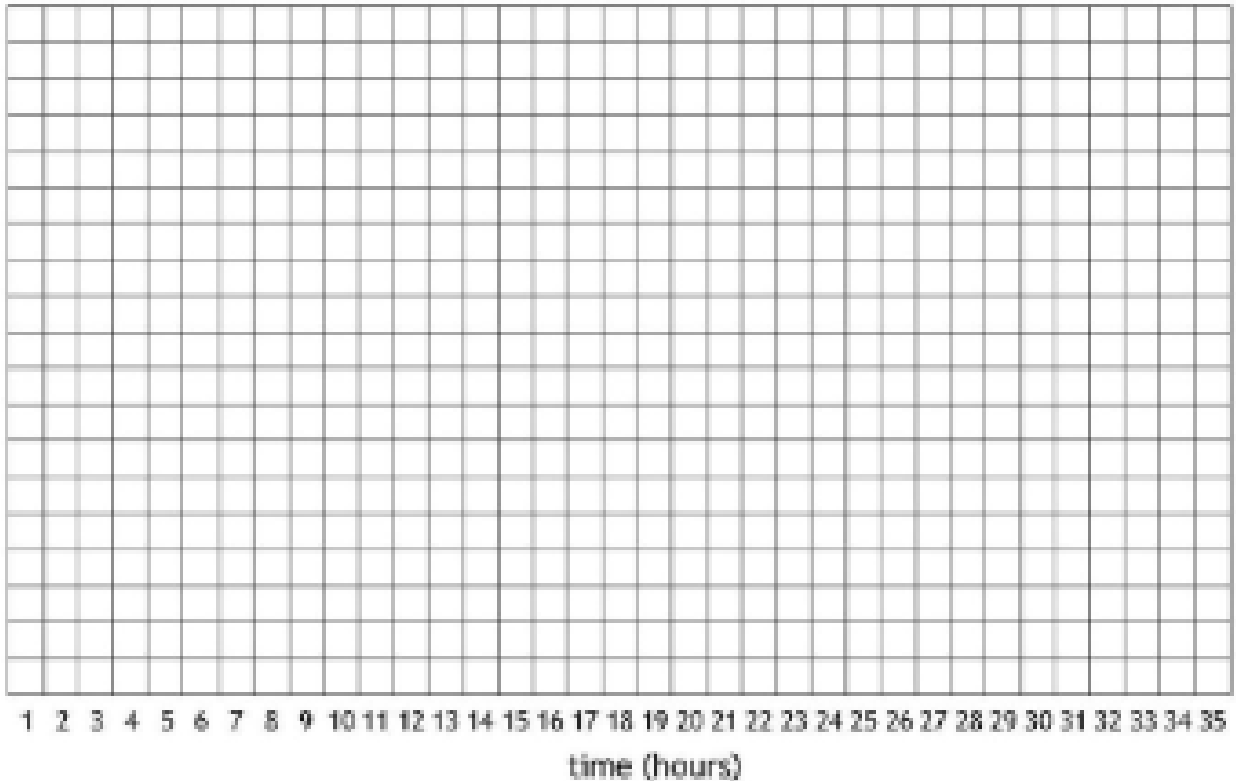
1

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(c) Construct a Gantt chart, without float times, for this job.

3



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VENN DIAGRAMS

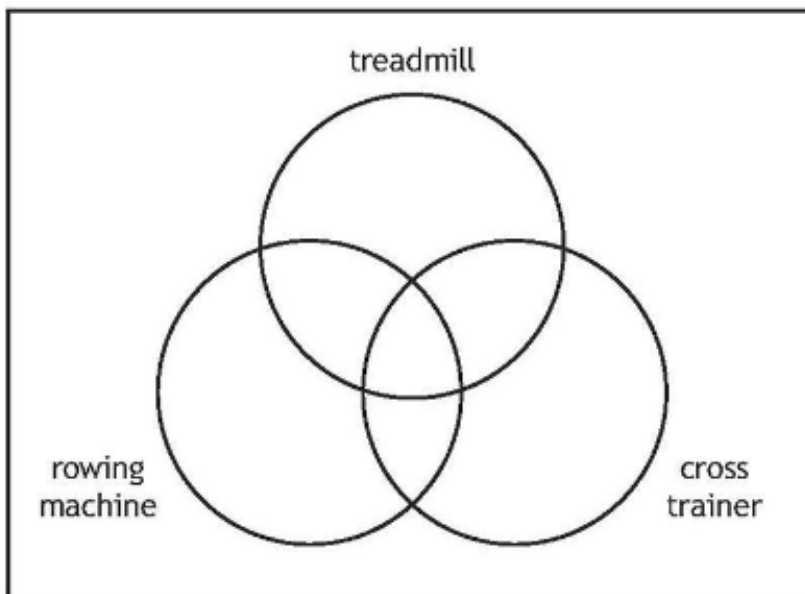
A group of gym members were asked what machines they regularly use from a choice of treadmill, rowing machine and cross trainer.

The results were as follows:

- 4 members use all three machines
- 38 members use both the treadmill and the rowing machine
- 32 members use both the cross trainer and the treadmill
- 15 members use both the rowing machine and the cross trainer
- 90 members use the treadmill
- 85 members use the rowing machine
- 53 members use the cross trainer
- 13 members use none of these machines.

(a) Complete the Venn diagram to show this information.

3



(b) A gym member is selected at random.

Determine the probability that they regularly use the rowing machine and the cross trainer, but not the treadmill.

2

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VENN DIAGRAMS

2. A college offers language courses in French, Spanish and Arabic.

A group of students are asked which, if any, of these language courses they study at this college:

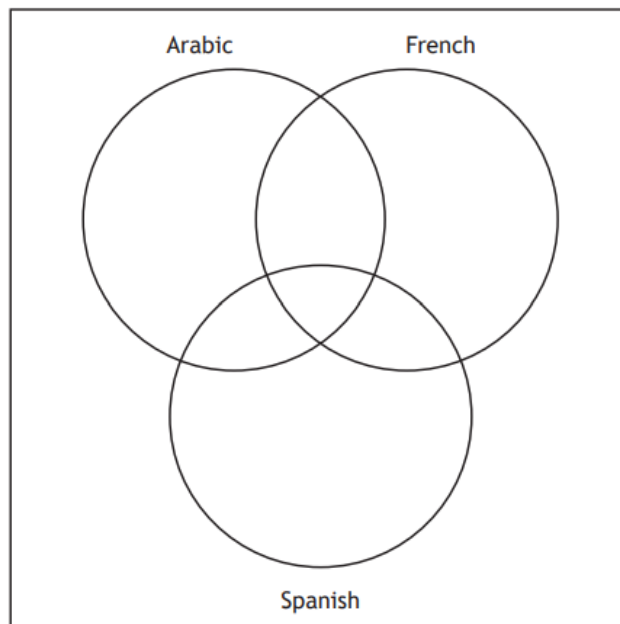
- 31 students study French, Spanish and Arabic
- 34 students study French only
- 7 students study Spanish only
- 13 students study Arabic only
- 51 students study French and Spanish
- 60 students study Spanish and Arabic
- 63 students study French and Arabic
- 14 students study no languages.

(a) Complete the Venn diagram to show this information.

3

(a) Complete the Venn diagram to show this information.

3



(b) A student is selected at random.

Determine the probability that the student studies Spanish and Arabic, but not French.

2

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PROBABILITY – EXPECTED COSTS

An events management company has been awarded a contract to construct the stages for a music festival.

If the construction of the stages is delayed the contract says that a penalty of £25,000 will be charged.

For the purposes of the cost-benefit analysis, the company has assumed only two events will cause a delay:

- a stage structure failing
- staff failing to attend the site.

The probability of **no delay** is 0.76.

(a) Calculate the expected cost of a delay.

2

The company is considering using the following control measures:

- Control Measure 1 — Pay another company to have staff on standby to attend the site in the case of their staff being unable to attend, at a cost of £1000.
- Control Measure 2 — Have a spare stage on standby in case of a failure, at a cost of £1500.

The probability of staff failing to attend the site is 0.2.

The probability of a stage structure failing is 0.05.

(b) Calculate the expected cost if control measure 2 is taken.

1

The expected cost if control measure 1 is taken is £2250.

The company will only use one control measure.

(c) Explain which control measure the company should use to minimise expected costs.

1

PROBABILITY – EXPECTED COSTS

An electric vehicle charging company has been awarded a contract to install 500 charging points in a city. As part of the contract a number of tasks must be completed including ordering and receiving charger parts and installing electric cables.

If the installation is delayed the company faces a fixed penalty of £75,000.

The company has identified the following two reasons for a possible delay:

- a shortage in parts for the chargers
- the electric cables will not be in place in time.

There is a 0.325 probability one or both of these delays will happen.

(a) Calculate the expected cost of a delay.

1

The company is considering using the following control measures:

- Control Measure 1 — Import extra parts at a cost of £15,000.
- Control Measure 2 — Pay another firm to help lay the cables, at an additional cost of £16,000.

There is a 0.1 probability of a shortage in parts for the chargers.

There is a 0.25 probability that the electric cables will not be in place in time.

(b) Calculate the expected cost of a delay using:

(i) only control measure 1

1

(ii) only control measure 2.

1

The company will only use one control measure.

(c) Based on your calculations, explain which control measure the company should use in order to minimise expected costs.

1

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HIGHER APPLICATIONS OF MATHS STATISTICS UNIT REVISION

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You must refer to the spreadsheet file 'Q5 Radio.csv' for the data, and the word processing file 'Q5 Radio Answers.docx' when answering this question.

You must complete parts (a) (i) and (a) (iii) using appropriate statistical software.

You must include all output from statistical software, and your answers in the word processing file 'Q5 Radio Answers.docx'.

A radio station asks their listeners to report how many hours they are listening for one week in July.

- (a) (i) Construct a histogram to summarise the data. 1
- (ii) Describe the distribution of the data shown in the histogram. 1
- (iii) Hence, generate and state the appropriate measure of location to summarise the data. 2

The data is then used to determine listening hours last summer.

- (b) State one reason why the data might misrepresent the listening hours last summer. 1

Print the word processing file 'Q5 Radio Answers.docx'.

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DOCUMENT
ANSWER FILE
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You must refer to the spreadsheet file 'Q6 golf.csv' for the data, and the word processing file 'Q6 Golf Answers.docx' when answering this question.

You must complete parts (a) (i) and (c) (ii) using appropriate statistical software.

You must include all output from statistical software, and your answers in the word processing file 'Q6 Golf Answers.docx'.

**SPREADSHEET
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A golf ball manufacturing company has designed a new golf ball which is more hard-wearing.

The company wants to determine whether there is a difference in the distances travelled by the current and new golf balls.

To measure the distances travelled by the current and new golf balls, a random sample of 40 of the current golf balls and a random sample of 40 of the new golf balls were subjected to distance tests using a mechanical hitting machine.

Data was collected for the distances travelled (in metres) of both the current and new golf balls.

The data was found to be approximately normally distributed.

**WORD
DOCUMENT
ANSWER FILE
LINK**

- | | | |
|-----|---|---|
| (a) | (i) Generate and state appropriate measures of location and spread for the current and new golf balls. | 2 |
| | (ii) Make two valid comparisons about the driving distances of both the current and new golf balls. | 2 |
| (b) | State appropriate null and alternative hypotheses to determine if there is a statistically significant difference between the distances travelled by the current and new golf balls. | 1 |
| | | |
| (c) | (i) State which type of hypothesis test is appropriate to determine if there is a statistically significant difference between the distances travelled by the current and new golf balls. | 1 |
| | (ii) Perform the hypothesis test and state the p -value. | 2 |
| | (iii) Hence interpret the p -value, and the result of the hypothesis test, in context. | 2 |

Print the word processing file 'Q6 Golf Answers.docx'.

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5. You must refer to the spreadsheet file 'Q5 Coffee.csv' for the data, and the word file 'Q5 Coffee Answers.docx' when answering this question.
- You must complete parts (a), (b) (i), and (c) (i) using appropriate statistical software.
- You must include all output from statistical software, and your answers in the word processing file 'Q5 Coffee answers.docx'.

A specialist coffee shop sells 22 different drinks.

For each drink the following data are provided:

- carbohydrate content (grams)
- number of calories (kCals).

The number of calories is dependent on the carbohydrate content of the drink.

- | | | |
|-----|---|---|
| (a) | Construct a scatter plot of number of calories on carbohydrate content. | 2 |
| (b) | (i) Find the correlation coefficient between number of calories and carbohydrate content. | 2 |
| | (ii) Interpret the correlation coefficient. | 1 |
| (c) | (i) Find the equation of the regression line of number of calories on carbohydrate content. | 2 |
| | (ii) Interpret the slope and intercept parameters. | 2 |
| (d) | Estimate the number of calories in a drink which has 59 g of carbohydrate. | 1 |

Print the word processing file 'Q5 Coffee Answers.docx'.

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ANSWER FILE
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You must refer to the information on 'Strength and conditioning' given in the pre-release material when answering this question.

You must also refer to the spreadsheet file 'Q7 Jump.csv' for the data, and the word processing file 'Q7 Jump Answers.docx' when answering this question.

You must complete parts (a) (i), (b) (i), (b) (ii), and (c) using appropriate statistical software.

You must include all output from statistical software, and your answers in the word processing file 'Q7 Jump Answers.docx'.

A strength and conditioning coach wants to increase vertical jump height performance in their trainees. The data in the spreadsheet file shows back squat weight (kg) and vertical jump height (cm).

- (a) (i) Construct a scatter plot of vertical jump height on back squat weight for the data. 2
- (ii) Make an appropriate comment about the relationship between vertical jump height and back squat weight. 1
- (b) (i) Find the correlation coefficient between back squat weight and vertical jump height. 2
- (ii) Find the equation of the regression line of vertical jump height on back squat weight. 2
- (c) Use your statistical software to estimate the vertical jump height for a trainee who can back squat 165 kg, and comment on the accuracy of the predicted value. 2

Based on the correlation, the coach advises the trainees that increasing their back squat weight will increase their vertical jump height.

- (d) Explain why the statistical analysis does not support this advice. 1

Print the word processing file 'Q7 Jump Answers.docx'.

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FILE LINK

WORD
DOCUMENT
ANSWER FILE

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2. Strength and conditioning

Strength and conditioning is an important part of overall health and fitness. A strength and conditioning coach uses a variety of different exercises to improve overall health and fitness.

Some of the exercises that a coach may use are:

- Back squat: a lower-body weight training exercise in which the trainee performs a squat using added weight. The back squat weight is usually recorded in kilograms.
- Bench press: an upper-body weight training exercise in which the trainee presses a weight upwards while lying on a weight training bench. The bench press weight is usually recorded in kilograms.
- Plank: a physical exercise designed to strengthen the abdominal muscles, in which the trainee performs a press-up and holds the raised position for a set period of time. Plank hold time is usually recorded in seconds.

One measurement of overall strength and condition is a person's vertical jump height. This is the height of a jump straight up into the air from a standing position. Vertical jump height is usually recorded in centimetres.

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You must refer to the spreadsheet file 'Q12 Eye Colour.xlsx' when answering this question.

You must complete parts (a), (b) and (c) using the spreadsheet file.

A random sample of 50 children were invited to take part in a general health study. They were asked to answer the following questions:

- State your sex.
Please select from: male, female, prefer not to say
- State your eye colour.
Please select from: blue, brown, green
- Are you colour blind?
Please select from: yes, no

The responses from the survey were partially entered into a 'Participant Responses' table.

Participants 3, 7, 13, and 18 responded 'yes' to being colour blind.

All other participants were not colour blind.

Participant 42 has withdrawn their consent for their responses to be included in the data.

Open the 'Eye Colour' worksheet.

- | | |
|--|---|
| (a) Complete the 'Participant Responses' table using the information above. | 2 |
| (b) Complete the 'Eye Colour Summary' table. | 2 |
| (c) Create an appropriate chart to display the frequencies in the 'Eye Colour Summary' data. | 3 |

Print the 'Eye Colour' worksheet in value view and formula view.

Ensure the chart is visible and contained on one page in each printout.

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CLASSIFYING DATA AND SURVEYS

3. An online survey is sent out by email to people on the electoral register who live in Renfrewshire.

The aim of the survey is to determine which political party people intend to vote for in an upcoming election.

The survey asks participants to answer the following questions:

- State your age in years.
- State your gender.
Please select from: male, female, non-binary, other, prefer not to say.
- Which political party do you intend to vote for in the upcoming election?
- Did you vote in the last election?
Please select from: yes, no, don't know, prefer not to say.
- If yes, state which political party you did vote for in the last election.

(a) State the type of data that best describes

(i) age

1

(ii) gender.

1

The results of the online survey are to be used to predict the outcome of an upcoming national election.

- (b) Give two reasons why these results do not provide a representative sample of the national population to allow the prediction to be accurate.

2

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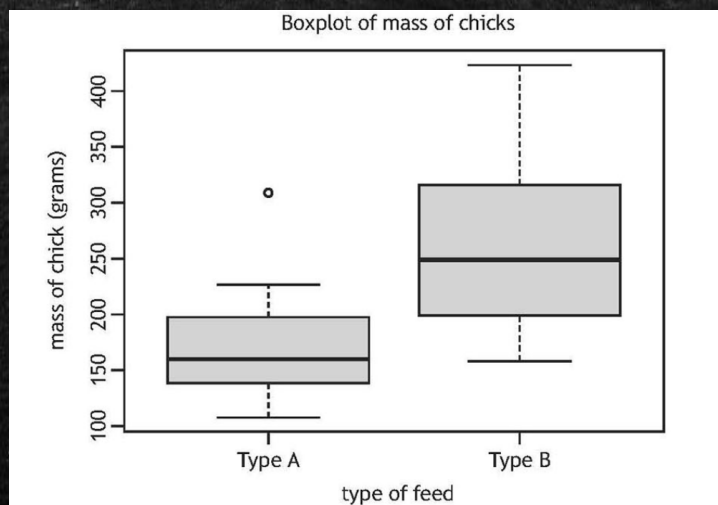
CLASSIFYING DATA – STATISTICAL TESTS

You must refer to the information on ‘Feeding chicks’ given in the pre-release material when answering this question.

To monitor the effect of the type of feed on their mass, a group of chicks were randomly allocated into two groups at birth.

One group was given feed type A and the second group was given feed type B from birth.

The comparative boxplot summarises the mass of chicks, in grams, when weighed at three weeks old.



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(a) Explain which type of feed results in chicks with the least variability in mass. 1

A hypothesis test was performed to compare the two groups.

(b) (i) State the hypothesis test performed. 1

(ii) Hence state the null and alternative hypotheses. 1

The p -value was 0.003147.

(iii) Interpret the p -value, and the result of the hypothesis test, in context. 2

CONTINUED ON NEXT PAGE

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CLASSIFYING DATA – STATISTICAL TESTS

The type of feed used for one group of chicks is sold in 4.5 kilogram bags.

The nutritional content is shown.

Nutritional content of feed	
Protein	720 g
Fat	150 g
Fibre	320 g
Other	3310 g

- (c) Explain whether this nutritional information is more likely to be for feed type A or type B.

1

3. Feeding chicks

One of the most important considerations for poultry owners is picking the right type of feed. When feeding freshly hatched chicks, it might be tempting to save money by feeding them the same feed given to adult birds.

However, a specially formulated diet that will help them grow at this early stage of life is required. Specialised chick feed is recommended from birth up until six weeks old. This type of feed has between 20% and 24% protein to boost their metabolism and enable them to increase their mass appropriately and feather out rapidly.

At three weeks old, chicks typically weigh 200 grams. If a chick has not been fed specialised chick feed, they are likely to experience stunted growth indicated by a mass of under 200 grams.

After six weeks, the chicks can slowly transition to another type of feed.

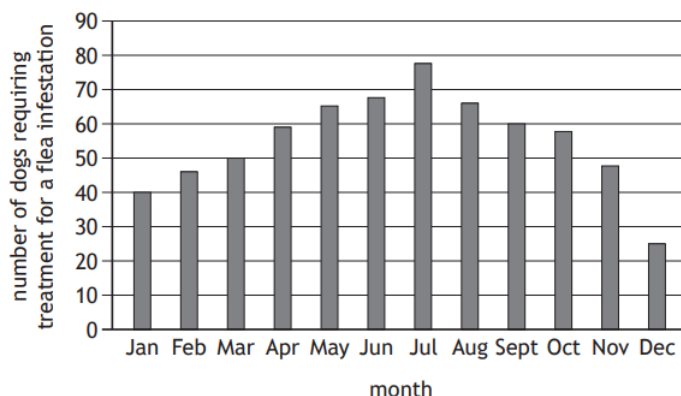
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CLASSIFYING DATA – STATISTICAL TESTS

3. You must refer to the information on 'Dogs and fleas' given in the pre-release material when answering this question.

After Study 1 the following bar chart was created. It shows the number of dogs requiring treatment for a flea infestation against the month of the year.



- (a) State the type of data that best describes the number of dogs requiring treatment for a flea infestation.

1

The student vet claims this graph proves that flea infestations are more common in the warmer months across the UK.

- (b) Explain why this claim is not valid.

1

The data gathered from Study 2 is shown in the table.

Area	Number of dogs inspected	Number of dogs with a flea infestation
Urban Clinic	213	30
Rural Clinic	156	43

- (c) State a statistical test that would be appropriate to determine whether this data provides evidence of a difference in flea infestations between urban and rural areas.

1

- (d) State one part of the design of Study 2 that would be needed to ensure that the assumptions required to perform the test in (c) are valid.

1

To fit in with the student vet's placements, the data from the urban clinic was collected from July to September, and the data from the rural clinic was collected from October to December.

- (e) Explain how this may affect the conclusions of Study 2.

1

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CLASSIFYING DATA – PRE RELEASE MATERIAL

1. Dogs and fleas

The UK is a nation of dog-lovers with 1 in 4 adults owning a dog. Owning a dog is a big commitment and it is important that dog owners meet the needs of their dogs and register them with a vet. One thing that dog owners need to be aware of is fleas.

Fleas are a blood-sucking parasite that are found in shaded areas near long grass and bushes. Fleas are found throughout the year and are believed to be more common in the warmer months. They are thought to be more common in rural areas. A dog is said to have a flea infestation when the number of fleas exceeds 20.

Study 1:

To investigate the seasonal variation of flea infestations in dogs, a student vet, on placement in a veterinary clinic in Edinburgh city centre, gathers data on the number of dogs requiring treatment for flea infestations. From appointment records, the number of treatments for flea infestations in dogs over a one-year period were recorded.

Study 2:

To investigate whether flea infestations are more common in rural areas than urban areas, a second study was designed by the student vet. In the study, a sample of dogs was inspected for flea infestations over a 3-month period at the Edinburgh clinic, and a similar sample of dogs was inspected for flea infestations at a rural veterinary clinic.

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HIGHER APPLICATIONS OF MATHS MODELLING unit Revision

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MATHEMATICAL MODELS

RECURRENCE RELATIONS (EXCEL)

You must refer to the information on 'Dracaena plants and atmospheric carbon dioxide levels' given in the pre-release material when answering this question.

You must also refer to the spreadsheet file 'Q4 Dracaena Plant.xlsx' when answering this question.

You must complete parts (a) and (b) using the spreadsheet file.

Part (c) must be completed in the answer space provided.

**DATABOOK
ON NEXT
PAGE!**

**LINK TO
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FILE**

A research scientist is studying the effect of Dracaena plants to improve indoor air quality in a kitchen showroom. They estimate that:

- large Dracaena plants reduce the concentration of carbon dioxide (CO_2) in the showroom by 13% each day
- each evening a heating system is left running. This adds enough CO_2 to increase the concentration in the showroom by 180 ppm.

At the start of the study the concentration of CO_2 in the showroom was 2000 ppm.

- (a) Complete the 'Dracaena plant study' worksheet to estimate the concentration of CO_2 at the end of 30 days. 3

The research scientist investigates the long-term concentration of CO_2 in the showroom.

- (b) Extend the table in your worksheet to the end of 60 days.

Construct a graph to show what will happen to the concentration of CO_2 in the showroom.

Your graph must include an appropriate title and axes labels. 3

- (c) Explain whether the large Dracaena plants are effective at obtaining very good indoor air quality in the showroom. 1

Print the 'Dracaena plant study' worksheet in value view and in formula view. Ensure the graph is positioned beside the table and is contained on one page in the printout.

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2. Dracaena plants and atmospheric carbon dioxide levels

Dracaena plants are a type of plant often grown indoors. They make excellent house plants because they tolerate low light and require little water. Dracaena plants have been shown to be effective air filters. They remove formaldehyde, benzene, trichloroethylene, and carbon dioxide from a room's atmosphere.

Carbon dioxide (CO_2) is a natural component of air, but in too large a concentration CO_2 can be harmful to humans. To improve indoor air quality many workplaces and schools take measures to reduce CO_2 concentration levels. Carbon dioxide concentration is measured in parts per million (ppm).

Acceptable CO_2 concentration levels vary from country to country. For example, in the UK, the Health and Safety Executive has a set of standards for acceptable CO_2 concentration levels in indoor spaces:

- 800 ppm or below indicates that an indoor space is likely to have very good indoor air quality.
- Consistent levels of 1500 ppm or above indicate that an indoor space is likely to have poor indoor air quality.

Where poor indoor air quality is identified, measures should be taken to improve ventilation — this will reduce the CO_2 concentration levels.

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MATHEMATICAL MODELS
RECURRENCE RELATIONS (EXCEL)



You must refer to the spreadsheet file 'Q8 Warehouse.xlsx' when answering this question.

You must complete parts (a) and (b) (i) using the spreadsheet file.

Parts (b) (ii), (c) and (d) must be completed in the answer space provided.

A warehouse company currently has 1750 units of stock. They deliver 20% of their current stock each week, and receive 300 units of new stock per week.

- (a) Complete the 'Warehouse Stock' worksheet to identify (in cell C10) the predicted number of units of stock in the warehouse at the end of week 26. 3
- (b) (i) Extend the table in your worksheet to construct a graph to show the units of stock for 52 weeks. 2
- (ii) Using the graph constructed in (b) (i), state which type of mathematical model best describes the units of stock in the warehouse. 1

After 52 weeks, the warehouse company plans to move to a smaller building with space for 1400 units of stock.

- (c) Comment on whether the smaller building will have enough space for the stock.
Justify your answer. 1
- (d) State one reason why this mathematical model may not be realistic. 1

Print the 'Warehouse Stock' worksheet in value view and formula view. Ensure the graph is positioned beside the table and is contained on one page in the printout.

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5. You must refer to the spreadsheet file 'Q5 School Roll.xlsx' when answering this question.

You must complete part (a) (i) and (c) (i) using the spreadsheet file.

Parts (a) (ii), (b), and (c) (ii) must be completed in the answer spaces provided.

A school is planning a new building as it is approaching its maximum capacity.

The school roll in August 2021 was 650 pupils.

- Approximately 18% of pupils leave by the end of each school year.
- 140 new S1 pupils join the roll in August each year.

(a) (i) Complete the 'School Roll' worksheet to predict the school roll in August 2031.

3

(ii) Comment on the precision of this prediction.

1

(b) Comment on the relationship between time and the predicted school roll up to August 2031.

1

The school moves forward with plans for a new building.

This will increase the capacity of the school to 800 pupils.

(c) (i) Extend the table in your worksheet to construct a graph to show what is predicted to happen to the school roll in the long term.

You must consider what happens to the school roll beyond August 2031.

2

(ii) Use your graph to determine whether the new capacity is suitable.

1

Print the 'School Roll' worksheet in value view and in formula view. Ensure the graph is positioned below the table and is contained on one page in the printout.

[LINK TO
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FILE](#)

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MODELLING – FERMI ESTIMATES

Estimate the total number of portions of fruit and vegetables that a typical person eats during their childhood.

State any assumptions you make.

3

1. A school year in Scotland usually lasts 190 days.

Estimate the total number of hours that a typical pupil in Scotland will spend in school during their lifetime.

State any further assumptions you make.

3

MODELLING – FERMI ESTIMATE & EXPONENTIAL GROWTH DATABOOK

10. You must refer to the information on ‘Mountain gorillas’ given in the pre-release material when answering this question.

The 2020 study found that the population of mountain gorillas had increased to 1004.

An expert has stated that if the mountain gorilla population in the Virunga Mountains continues to grow exponentially there will be 1600 gorillas by the year 2032.

- (a) Determine if the expert’s statement is correct.

Give a reason for your answer.

2

A typical adult mountain gorilla eats 30 kg of food per day.

- (b) Estimate the **maximum** amount of termites and ants (in kg) that a typical mountain gorilla will eat during their adult lifetime.

State any assumptions you have made.

3

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4. Mountain gorillas

Deforestation is happening on a massive scale throughout the world and many animals are experiencing habitat loss. Conservation teams have been campaigning against deforestation in a bid to increase wildlife populations.

One success story is the mountain gorilla. More than half of the world's mountain gorillas live in the Virunga Mountains, a range of extinct volcanoes that border the Democratic Republic of Congo, Rwanda and Uganda.

A study in 2008 showed there were only 680 mountain gorillas in the Virunga Mountains. A further study in 2020 showed that due to conservation efforts, including clamping down on illegal logging, the population of mountain gorillas had increased.

Mountain gorillas are considered adults when they are 10 years old and will live for an average of 35 years.

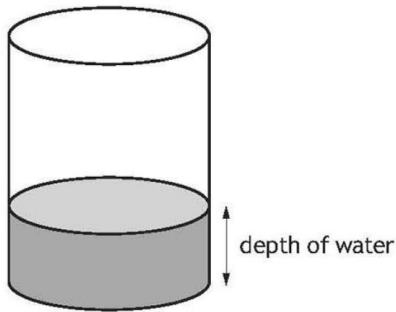
Mountain gorillas survive on a **mainly** vegetarian diet consisting of bamboo shoots, fruit, and berries. Given the opportunity, mountain gorillas will also eat ants and termites, but their diet will remain mostly vegetarian.

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MODELLING – RELATIONSHIP FROM GRAPH

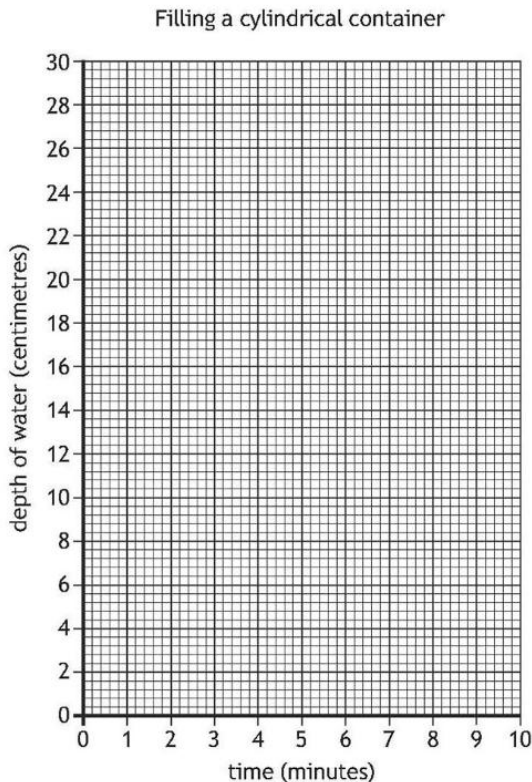
A cylindrical container is being filled with water.



- Initially, the depth of water is 4 centimetres.
- The depth of the water in the container increases at a constant rate of 3 centimetres per minute.
- It takes 8 minutes to finish filling the container.

(a) Draw a graph to model how the depth of water in the container changes over time until it is filled. Use the axes provided.

3



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MODELLING - RELATIONSHIP FROM GRAPH

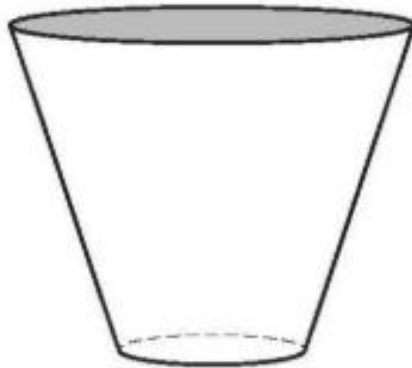
(b) (i) State the type of relationship modelled in your graph.

1

(ii) State the dependent variable in your graph.

1

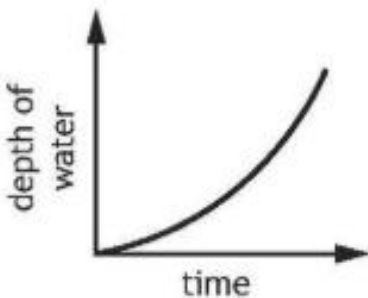
Another container is shown below.



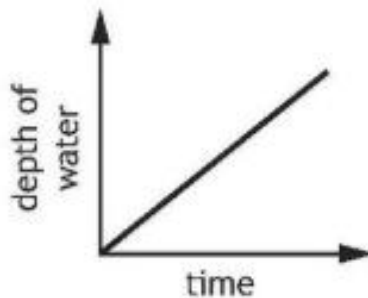
This container is also filled with water at a constant rate of 5 cubic centimetres per second.

Three graphs are shown below showing how the depth of water in the container changes over time.

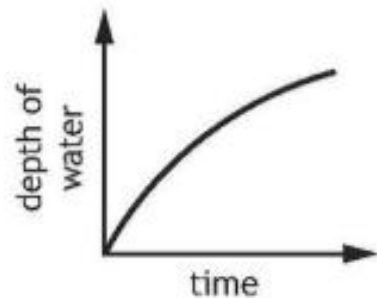
Graph A



Graph B



Graph C



(c) Explain which graph could model the depth of water in the container.

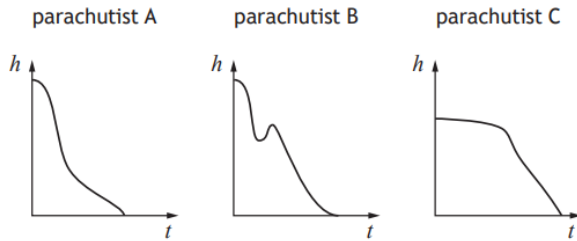
2

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MODELLING – RELATIONSHIP FROM GRAPH

6. The three diagrams, below, show how parachutists' heights vary above the ground over a period of time.



State which graph could not model their jump.

Explain your answer clearly.

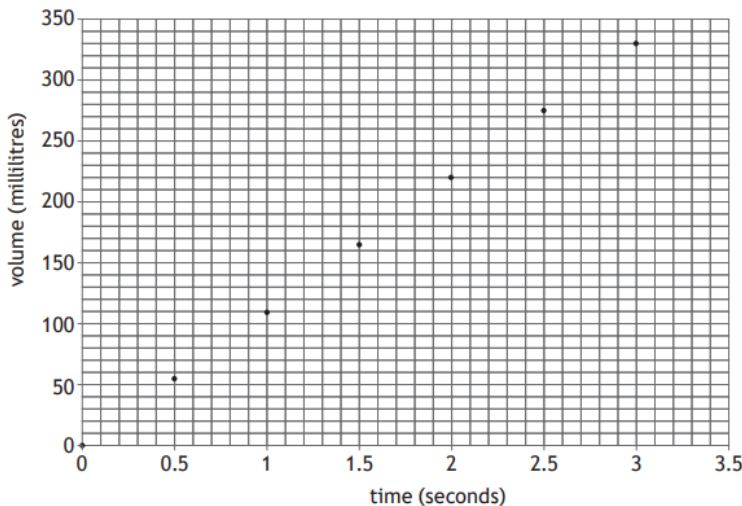
2

10. A factory fills cans with a carbonated fruit drink.

The cans produced have a volume of 330 millilitres.

The graph below shows the relationship between time, in seconds, and volume of carbonated fruit drink, in millilitres, as a can is filled.

Scatterplot of volume of carbonated fruit drink against time



- (a) State the type of relationship modelled in the graph.

1

- (b) Determine the rate at which each can is filled with carbonated fruit drink.

Your answer must include appropriate units.

2

The factory can only fill the cans when there are staff present.

The machines in the factory can fill 5 cans at the same time.

- (c) Estimate how many cans the factory can fill in one week.

State any assumptions you have made.

3

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MODELLING – DATABOOK & USING A FORMULA/GRAPH

10. You must refer to the information on 'E10 Petrol' given in the pre-release material when answering this question.

At the end of June 2021, the UK government replaced the standard E5 fuel with E10 in a bid to reduce national CO₂ emissions.

A newspaper article stated that:

'Since the nationwide introduction of E10 fuel, CO₂ emissions have reduced to the effect of taking 550 000 cars off the road by the end of 2022.'

- (a) (i) Calculate the CO₂ emissions of 550 000 cars.

1

- (ii) Suggest a reason why the information from the newspaper article might not be appropriate as part of any further research on reducing CO₂ emissions.

1

The fuel economy of a car, F miles per gallon (mpg), is reduced when it carries an additional load of m kilograms.

The fuel economy of a small car is modelled using the following equation:

$$F = 73.6 \times 0.98^{\frac{m}{45}}$$

- (b) (i) State the independent variable.

1

- (ii) Estimate the fuel economy of the small car when it is carrying an additional load of 150 kg.

1

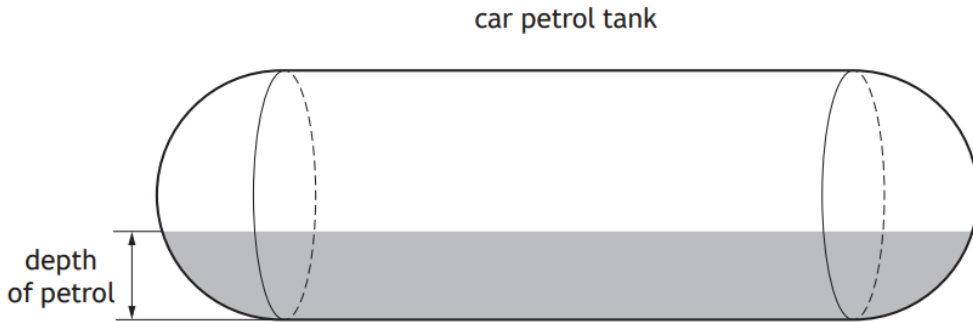
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A customer is filling up their tank with petrol.

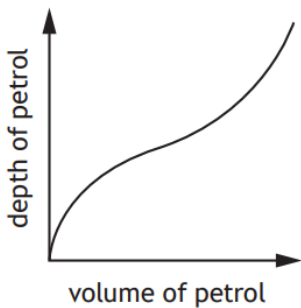
The shape of the petrol tank inside the car is a horizontal cylinder with two hemispheres on either side as shown in the diagram.



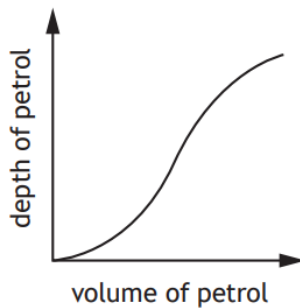
The three diagrams below show how the depth of petrol varies with the volume of petrol in the tank.

The three diagrams below show how the depth of petrol varies with the volume of petrol in the tank.

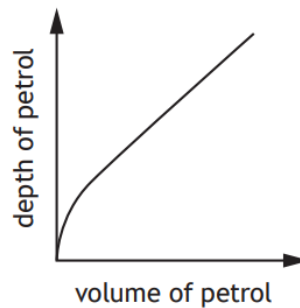
Graph A



Graph B



Graph C



(c) Explain which graph could model the depth of petrol in the tank.

2

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2. E10 Petrol

The following information is taken from the government's website
<https://www.gov.uk/guidance/e10-petrol-explained>

Petrol in the UK currently contains 5% renewable ethanol (known as E5) or 10% renewable ethanol (known as E10).

Reducing emissions

CO₂ is one of the greenhouse gases that contribute to climate change and the main benefit of E10 petrol is that it reduces overall levels of CO₂-based vehicle emissions.

By blending petrol with up to 10% renewable ethanol, less fossil fuel is needed, helping us reduce carbon emissions and meet climate change targets.

The introduction of E10 petrol in the UK is predicted to cut transport CO₂ emissions by 750 000 tonnes a year — the equivalent of taking 350 000 cars off the road.

Renewable fuel blends, such as E10 petrol, are generally introduced to reduce overall CO₂ emissions. They have little impact on emissions associated with air quality and public health.

The production of renewable ethanol for blending with fossil petrol also results in valuable by-products, including animal feed and stored CO₂.

Fuel economy

Using E10 petrol can slightly reduce fuel economy (the number of miles you are able to drive on a gallon of fuel). You may see a reduction of around 1%, but it is unlikely to be noticeable in everyday driving.

Other factors, such as your driving style or driving with under-inflated tyres or a roof rack, have a much more significant impact on fuel economy than using E10 petrol.

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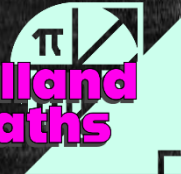
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